Experiment-4

Capture and Analyze network traffic using wireshark

Aim:

To capture, save and analyze network traffic including TCP, UDP, IP, HTTP, ARP Protocols using wireshark network analyzer tools.

Requirements:

wireshark 4.x installed, Linux or Windows Administrative Privileges, active network connection, web browser.

Procedure:

1) Install wireshark: Sudo apt install wireshark.

2) Launch wireshark with admin Privileges.

3) Select the active network interface.

4) Start capture by clicking the wireshark icon button

commands:

install wireshark on Linux

sudo apt-get install wireshark -y

sudo usermod -aG wireshark $USER

Capture with Tshark

sudo tshark -i eth0 -c 100 -w capture.pcap

display filters

tcp # All TCP traffic

udp # All UDP traffic

http # HTTP only

dns # dns queries / responses

arp # ARP traffic

icmp # Ping traffic

dhcp # DHCP DORA Process

tshark -r capture.pcap -Y 'dns.qry.name' -T fields

tshark -r capture.pcap -q -z io,phs

Protocol hierarchy

5) Generate traffic: open a browser, ping a host, run nslookup.

6) Stop capture after 30 seconds.

7) Filter DHCP: type 'dhcp' and examine DORA process.

8) Filter ARP: type 'arp' and view MAC-IP resolution.

9) Export the capture file as PCAP.

10) Use Statistics → Protocol Hierarchy to see traffic distribution.

Result:

Network traffic was successfully captured and analysed using Wireshark. TCP handshake, DNS queries, HTTP requests, ARP resolutions, DHCP lease process and ICMP exchanges were identified and examined.

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